1) # of MACs

MACs = input dimension · output dimensions

Layer 1: $784 \cdot 256 = 200704$

Layer 2: $256 \cdot 128 = 32768$

Layer 3: $128 \cdot 10 = 1280$

2) Total MACs = 234752 for one forward pass

3) total of trainable parameters inc biases

Weight = input dimensions · output dimensions

Total parameters = <u>234,752</u>

4) weight memory = 234,752 · 4 bits

= 939,008 bytes

5) total activition memory in bytes:

$784 + 256 + 128 + 10 = 1,178$

$1,178 \cdot 4$ bytes = 4,712 bytes

6) Arithmetic Intensity = $\dfrac{2 \cdot \text{total MACs}}{\text{weight byte} + \text{activation mem}}$

$= \dfrac{2 \cdot 234752}{939,008 + 4,712}$

$= 0.498$